Rajveer Gupta

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Summary

AI engineer who builds **LLM agents that take actions, not just chat**: multi-agent LangGraph workflows with human handoff, grounded RAG over HR policies, and LLM agents served via FastAPI. Seeking an **Agentic AI internship**.

Education

Indian Institute of Information Technology, Nagpur Aug 2024 – Jun 2028
B.Tech, Computer Science & Engineering — CGPA: 9.08/10 *Nagpur, India*
Coursework: DSA, OOP (AA), Software Engineering (AA), Operating Systems (AB), DBMS, Programming Language Design (AB)

Achievements

- **AIR 471 of 32K teams** — Amazon ML Challenge 2026: led a team of 4 that matched 11.7M records to real businesses.
- **Top 3,000 of 134,000+** — selected for Amazon ML Summer School 2026, taught by Amazon scientists.
- **Best Innovation Award** — Tech Expo 2026, Tinkerers' Lab, IIT Hyderabad, for an AI assistant for Alzheimer's patients.
- **MVP Round, Microsoft Imagine Cup 2025** — international round, with the same Alzheimer's assistant.
- **Finalist, IEEE IES Generative AI Hackathon 2026** — the idea grew into the IEEE IRAI 2026 paper below.

Projects

ReturnGuard — Multi-Agent Return-Fraud Detection with Human Handoff 2026
GitHub / Demo — Python, LangGraph, LangChain, Gemini 2.5 Flash, DINOv2, Flask, SQLite
What it does: checks if a returned product is the one delivered, then accepts it, rejects it, or **hands it to a human** with reasons.

- **Architected the full agent graph** in LangGraph: shared typed state, conditional routing, and a reusable vision-language **subgraph** run **in parallel** per photo view (fan-out / fan-in) before an LLM judge.
- **Cut LLM cost with a two-tier design**: DINOv2 embeddings auto-clear near-identical items; the LLM reviews the rest.
- Logged each case's **0–100 risk score**, reasoning and verdict in SQLite (**parameterized queries**) behind Flask REST APIs.
- **Tuned for precision so honest customers are not refused**: caught DINOv2 rating *different* same-material products 85–92% similar and tightened the threshold. **0.93 precision / 0.88 F1**.

Company Policy RAG Assistant — Grounded Q&A over HR Documents 2026
GitHub — LangChain, LLaMA, ChromaDB, Hugging Face embeddings
What it does: employees ask about leave or reimbursements in plain English and get answers taken only from the real policies.

- Rebuilt a DPR + GPT-2 + FAISS prototype as a **LangChain RetrievalQA** pipeline with a persisted ChromaDB store.
- Added an **anti-hallucination prompt**: says "I don't know" when no policy covers it, refuses off-topic queries, cites sources.

MediGuard-AI — LLM Agent for Context-Aware ICU Alerts 2026
Live / GitHub — Python, Qwen2.5-72B, OpenEnv, FastAPI, Gradio, Docker · Team of 3, Meta PyTorch × Scaler Hackathon
What it does: an agent decides if a patient's vitals need no action, a nurse check or emergency alert, using context to cut false alarms.

- Co-built an **LLM agent** (Qwen2.5-72B via an OpenAI-compatible API) with a rule-based fallback and structured reasoning logs; beat rule-based baselines by **+0.07 / +0.22 / +0.23** across three tasks.
- Cut episodes from 360 to 60 steps to fit a 30-minute eval budget; served via FastAPI + Gradio in Docker on HF Spaces.

Experience

Research Intern, Computer Vision — IIIT Nagpur (Dr. Nishat Afshan Ansari) 2026
What it does: spots which factory camera has gone bad (blurred, noisy, blocked or dead) and rebuilds its view from nearby cameras.

- **Detect:** monitors every feed frame by frame; **hysteresis-based fault classification** flags a camera only when the fault persists, so one glitchy frame does not raise a false alarm.
- **Reconstruct and restore:** rebuilds a dead camera's view from its neighbors with **3D Gaussian reconstruction**, fills unseen areas with diffusion, and cleans blurred frames with a Mamba-CNN that runs in linear, not quadratic, time.
- **Results:** PSNR of 30.02 dB (Wildtrack) and **32.42 dB** (GoPro), beating NAFNet and MoCE-IR; ablations per module.

Publications

A. Gupta, A. Singh, **R. Gupta**, P. Jain, T. Patidar, "Fault Tolerant Industrial Vision Framework Through Semantic-Guided Cross-View Temporal Reconstruction and Restoration," *IEEE IRAI 2026*, Melbourne, Australia (accepted and presented).

Skills

Languages: Python, C++, SQL, Java, JavaScript
LLMs & Agents: LangGraph, LangChain, multi-agent orchestration, RAG, OpenAI / Gemini APIs, prompt engineering
Backend & Data: FastAPI, Flask, REST APIs, SQLite, PostgreSQL, MySQL, ChromaDB, FAISS, Docker, Git, Linux
ML: PyTorch, scikit-learn, Hugging Face Transformers, DINOv2, BERT, LightGBM